

# PHANTOM on GPUs with OpenMP

Thomas Bending



University  
of Exeter

IA  
SR & HR

NASA, ESA, and the Hubble Heritage Team (STScI/AURA)

NASA and The Hubble Heritage Team (STScI/AURA)

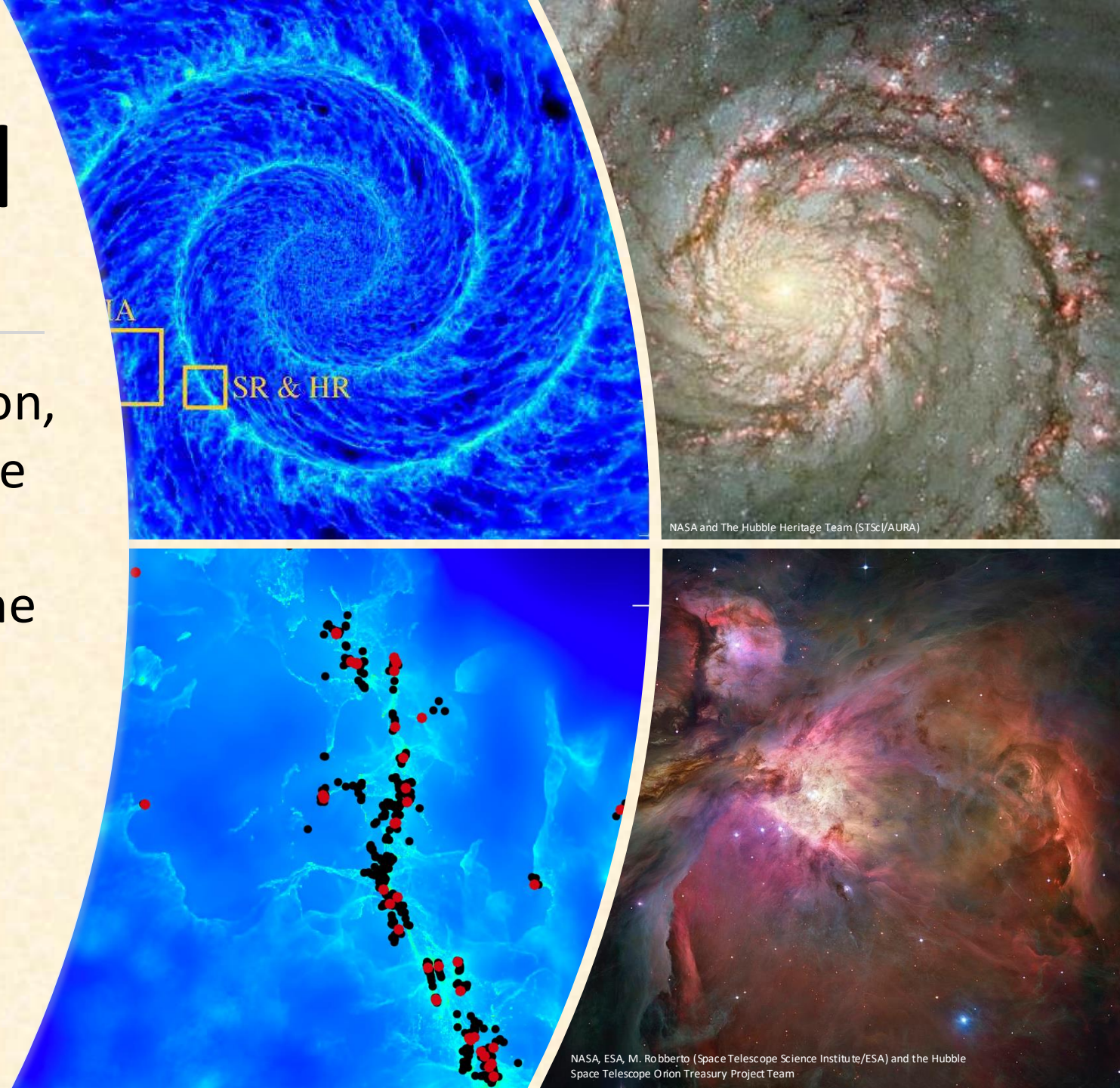
NASA, ESA, J. Hester and A. Loll (Arizona State University)

NASA, ESA, M. Robberto (Space Telescope Science Institute/ESA) and the Hubble  
Space Telescope Orion Treasury Project Team



# My background

- Using SPH to study star formation, stellar feedback, the ISM and the IMF.
- Primarily used **sphNG**, with some experience using **PHANTOM**.
- Increasingly focused on code optimization through algorithm redesign, MPI strategy and recently GPUs.



# GPU project

- **Implemented Morton-ordered (z-order) octree.**
- **Avoided neighbour searches outside tree build; all neighbour summations vectorised.**
- Attended DiRAC AMD hackathon with Bec Nealon (Warwick) and Adam Koval (Edinburgh); ported and ran code on AMD MI300A.
- Integrated CPU code into PHANTOM for density solving, matching original code speed.
- Most recently have the code running on NVIDIA A100.

# Morton order (z-order) bit interleaving

| $y \backslash x$ | 00   | 01   | 10   | 11   |
|------------------|------|------|------|------|
| 00               | 0000 | 0001 | 0100 | 0101 |
| 01               | 0010 | 0011 | 0110 | 0111 |
| 10               | 1000 | 1001 | 1100 | 1101 |
| 11               | 1010 | 1011 | 1110 | 1111 |

(a)

| $y \backslash x$ | 0  | 1  | 2  | 3  |
|------------------|----|----|----|----|
| 0                | 0  | 1  | 4  | 5  |
| 1                | 2  | 3  | 6  | 7  |
| 2                | 8  | 9  | 12 | 13 |
| 3                | 10 | 11 | 14 | 15 |

(b)



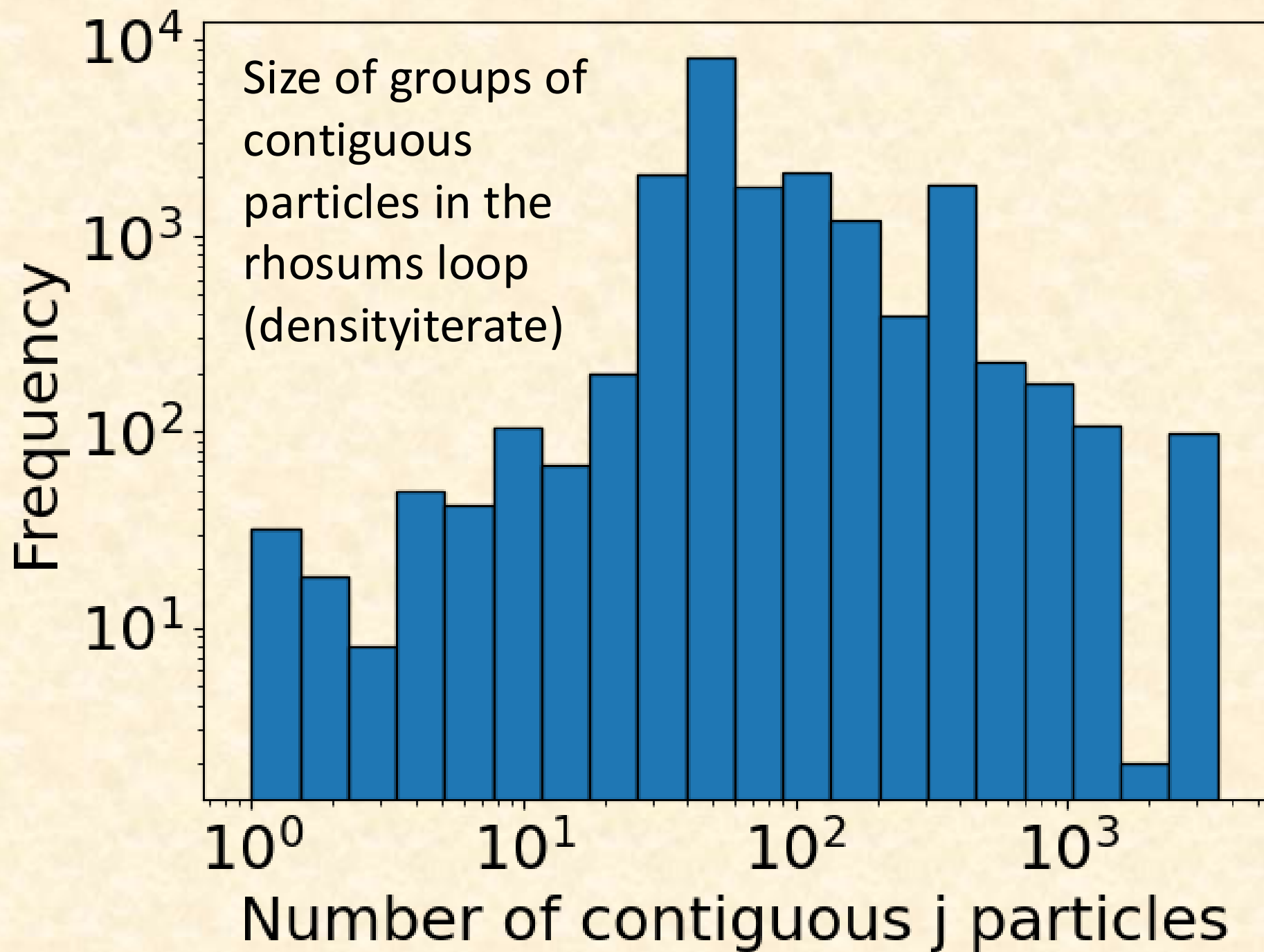
# Morton order (z-order) bit interleaving

| $y \backslash x$ | 00   | 01   | 10   | 11   |
|------------------|------|------|------|------|
| 00               | 0000 | 0001 | 0100 | 0101 |
| 01               | 0010 | 0011 | 0110 | 0111 |
| 10               | 1000 | 1001 | 1100 | 1101 |
| 11               | 1010 | 1011 | 1110 | 1111 |

(a)

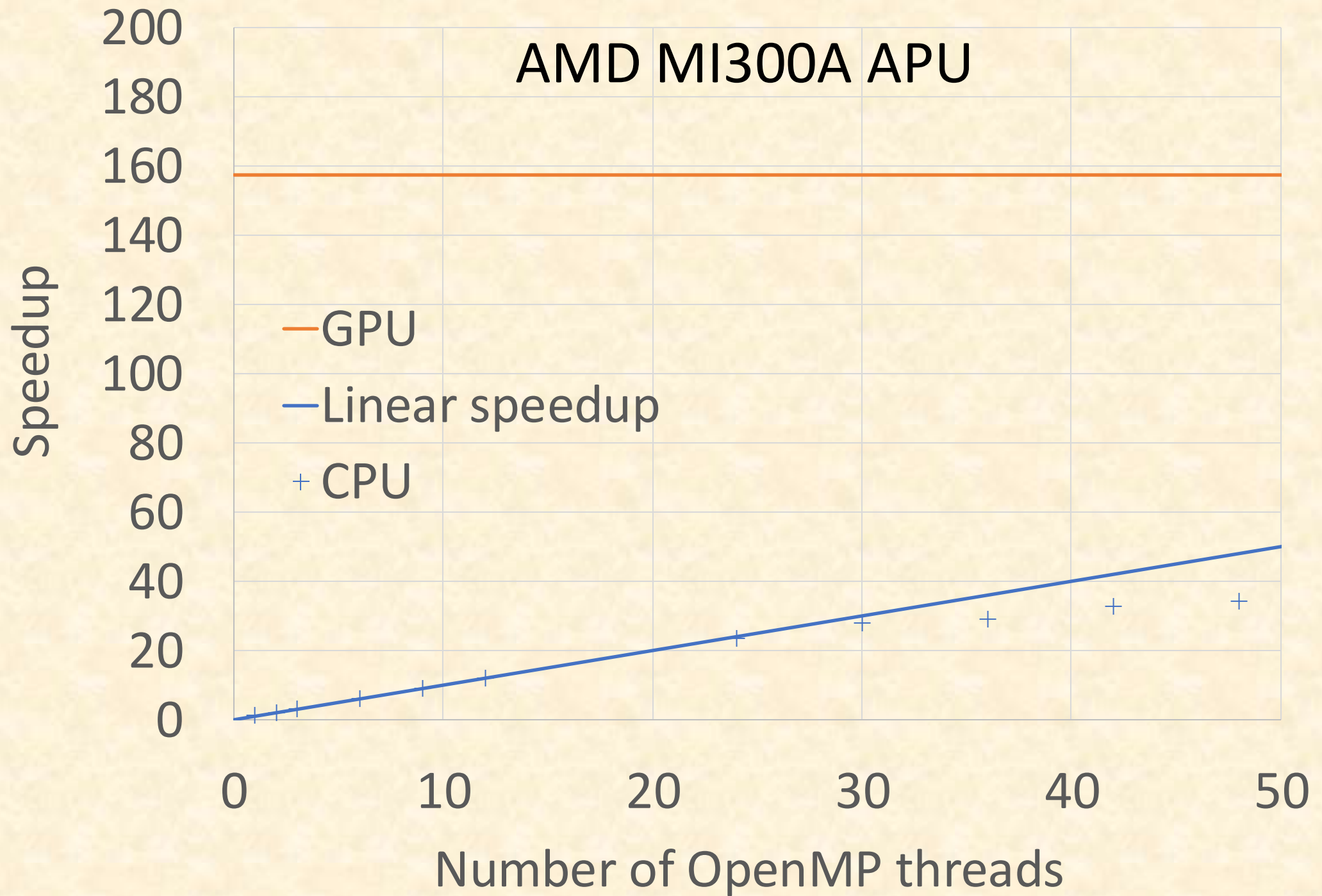
| $y \backslash x$ | 0  | 1  | 2  | 3  |
|------------------|----|----|----|----|
| 0                | 0  | 1  | 4  | 5  |
| 1                | 2  | 3  | 6  | 7  |
| 2                | 8  | 9  | 12 | 13 |
| 3                | 10 | 11 | 14 | 15 |

(b)



# GPU project

- Implemented Morton-ordered (z-order) octree for density solving on CPUs.
- Avoid neighbour searches and lists outside tree build; all neighbour summations vectorised.
- **Attended DiRAC AMD hackathon with Bec Nealon (Warwick) and Adam Koval (Edinburgh); ported code and ran on AMD MI300A.**
- Integrated CPU code into PHANTOM for density solving, matching original code speed.
- Most recently have the code running on NVIDIA A100.





# GPU project

- Implemented Morton-ordered (z-order) octree for density solving on CPUs.
- Avoid neighbour searches and lists outside tree build; all neighbour summations vectorised.
- Attended DiRAC AMD hackathon with Bec Nealon (Warwick) and Adam Koval (Edinburgh); ported and ran code on AMD MI300A.
- **Integrated CPU code into PHANTOM for density solving, matching original code speed.**
- **Most recently have the code running on NVIDIA A100.**

# Possible objectives for this week

- Run code on new hardware.
- Explore MPI strategy options.
- Enable iterations for GPU density solve.
- Handle neighbours outside zone of compact support.
- Implement GPU density solve in PHANTOM.
- **Connect with interested collaborators.**